# Exercise Sheet 3 (theory part): solution

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## Exercise 1: Principal Component Analysis (15 + 15 P)

We consider a dataset $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N \in \mathbb{R}^d$. Principal component analysis searches for a unit vector $\boldsymbol{u} \in \mathbb{R}^d$ such that projecting the data on that vector produces a distribution with maximum variance. Such vector can be found by solving the optimization problem:

$$\arg\max_{\boldsymbol{u}} \ \frac{1}{N}\sum_{i=1}^{N} \left(\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})\right)^2 \quad \text{with} \quad \|\boldsymbol{u}\|^2 = 1 \quad \text{and} \quad \boldsymbol{m} = \frac{1}{N}\sum_{i=1}^{N} \boldsymbol{x}_i.$$

**(a)** Show that the problem above can be rewritten as

$$\arg\max_{\boldsymbol{u}} \ \boldsymbol{u}^\top \Sigma\, \boldsymbol{u} \quad \text{with} \quad \|\boldsymbol{u}\|^2 = 1, \qquad \text{where} \quad \Sigma = \frac{1}{N}\sum_{i=1}^{N} (\boldsymbol{x}_i - \boldsymbol{m})(\boldsymbol{x}_i - \boldsymbol{m})^\top$$

is the covariance matrix.

**Solution (a):** Since $\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})$ is a scalar, it equals its own transpose, so its square can be written as a product of two copies:

$$\left(\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})\right)^2 = \boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})\,(\boldsymbol{x}_i - \boldsymbol{m})^\top \boldsymbol{u}.$$

Averaging over the data and pulling the constant vectors $\boldsymbol{u}$ out of the sum (linearity):

$$\frac{1}{N}\sum_{i=1}^{N} \left(\boldsymbol{u}^\top(\boldsymbol{x}_i - \boldsymbol{m})\right)^2 = \boldsymbol{u}^\top \underbrace{\left(\frac{1}{N}\sum_{i=1}^{N} (\boldsymbol{x}_i - \boldsymbol{m})(\boldsymbol{x}_i - \boldsymbol{m})^\top\right)}_{\Sigma} \boldsymbol{u} = \boldsymbol{u}^\top \Sigma\, \boldsymbol{u}.$$

The constraint $\|\boldsymbol{u}\|^2 = 1$ is unchanged, hence the problem is $\arg\max_{\boldsymbol{u}} \boldsymbol{u}^\top \Sigma \boldsymbol{u}$ s.t. $\|\boldsymbol{u}\|^2 = 1$.

**(b)** Show using the method of Lagrange multipliers that the problem above can be reformulated as solving the eigenvalue problem $\Sigma\boldsymbol{u} = \lambda\boldsymbol{u}$ and retaining the eigenvector $\boldsymbol{u}$ associated to the highest eigenvalue $\lambda$.

**Solution (b):** Lagrangian (with multiplier $\lambda$ for the constraint):

$$\mathcal{L}(\boldsymbol{u}, \lambda) = \boldsymbol{u}^\top \Sigma \boldsymbol{u} + \lambda\,(1 - \|\boldsymbol{u}\|^2).$$

Stationarity in $\boldsymbol{u}$ (using that $\Sigma$ is symmetric):

$$\frac{\partial \mathcal{L}}{\partial \boldsymbol{u}} = 2\Sigma\boldsymbol{u} - 2\lambda\boldsymbol{u} = 0 \quad \Longrightarrow \quad \Sigma\boldsymbol{u} = \lambda\boldsymbol{u}.$$

Hence every stationary point $\boldsymbol{u}$ is an eigenvector of $\Sigma$. To decide which one, multiply both sides by $\boldsymbol{u}^\top$ and use $\boldsymbol{u}^\top\boldsymbol{u} = 1$:

$$\boldsymbol{u}^\top \Sigma \boldsymbol{u} = \boldsymbol{u}^\top(\lambda\boldsymbol{u}) = \lambda.$$

The left-hand side is the objective, the right-hand side is the eigenvalue. Maximizing the objective therefore means choosing the eigenvector associated with the largest eigenvalue. ■

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## Exercise 2: Bounds on Eigenvalues (10 + 10 P)

Let $\lambda_1$ denote the largest eigenvalue of the matrix $\Sigma$. The eigenvalue $\lambda_1$ measures the variance of the data when projected on the first principal component. We study how the latter can be bounded with the diagonal elements of the matrix $\Sigma$.

*Preliminary observation* for all eigenvalues/eigenvectors $(\lambda_i, \boldsymbol{u}_i)$: show that $\lambda_i = \boldsymbol{u}_i^\top \Sigma \boldsymbol{u}_i \geq 0$.

**Solution (preliminary observation):** For an eigenpair $(\lambda_i, \boldsymbol{u}_i)$ with $\|\boldsymbol{u}_i\| = 1$:

$$\lambda_i = \boldsymbol{u}_i^\top \Sigma \boldsymbol{u}_i = \frac{1}{N} \sum_{k=1}^{N} \left(\boldsymbol{u}_i^\top (\boldsymbol{x}_k - \boldsymbol{m})\right)^2 \;\geq\; 0,$$

i.e. $\Sigma$ is positive semi-definite and all eigenvalues are non-negative.

**(a)** Show that $\sum_{i=1}^{d} \Sigma_{ii}$ is an upper bound to the eigenvalue $\lambda_1$.

**Solution (a):** The trace of a matrix equals the sum of its eigenvalues. Since all $\lambda_i \geq 0$ and $\lambda_1$ is one of the terms:

$$\sum_{i=1}^{d} \Sigma_{ii} = \operatorname{tr}(\Sigma) = \sum_{i=1}^{d} \lambda_i \;\geq\; \lambda_1.$$

**(b)** Show that $\max_{i=1}^{d} \Sigma_{ii}$ is a lower bound to the eigenvalue $\lambda_1$.

**Solution (b):** The canonical basis vectors $\boldsymbol{e}_i$ are unit vectors, so they are feasible points of the maximization defining $\lambda_1$. Maximizing over a subset can only give a smaller value:

$$\lambda_1 = \max_{\|\boldsymbol{u}\|=1} \boldsymbol{u}^\top \Sigma \boldsymbol{u} \;\geq\; \max_{\boldsymbol{u} \in \{\boldsymbol{e}_1, \ldots, \boldsymbol{e}_d\}} \boldsymbol{u}^\top \Sigma \boldsymbol{u} = \max_{i=1}^{d} \boldsymbol{e}_i^\top \Sigma \boldsymbol{e}_i = \max_{i=1}^{d} \Sigma_{ii}.$$

Together: $\max_i \Sigma_{ii} \;\leq\; \lambda_1 \;\leq\; \sum_i \Sigma_{ii}$. ■

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## Exercise 3: Iterative PCA (5 + 10 + 5 P)

When performing principal component analysis, computing the full eigendecomposition of the covariance matrix $\Sigma$ is typically slow, and we are often only interested in the first principal components. An efficient procedure to find the first principal component is *power iteration*. It starts with a random unit vector $\boldsymbol{w}^{(0)} \in \mathbb{R}^d$, and iteratively applies the parameter update

$$\boldsymbol{w}^{(t+1)} = \frac{\Sigma \boldsymbol{w}^{(t)}}{\|\Sigma \boldsymbol{w}^{(t)}\|}$$

until some convergence criterion is met. Here, we would like to show the exponential convergence of power iteration. For this, we look at the error terms

$$E_k(\boldsymbol{w}) = \frac{\boldsymbol{w}^\top \boldsymbol{u}_k}{\boldsymbol{w}^\top \boldsymbol{u}_1} \qquad \text{with } k = 2, \ldots, d,$$

and demonstrate that they all converge to zero as $\boldsymbol{w}$ approaches the eigenvector $\boldsymbol{u}_1$ and becomes orthogonal to the other eigenvectors. Throughout, $\Sigma \boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i$ with $\lambda_1 > |\lambda_k|$ for $k \geq 2$ (dominant eigenvalue).

**(a)** Show that $E_k(\boldsymbol{w}^{(t+1)}) = \dfrac{\boldsymbol{w}^{(t)\top} \Sigma \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top} \Sigma \boldsymbol{u}_1}$.

**Solution (a):** Insert the update rule into the definition of the error. The scalar normalization $1/\|\Sigma \boldsymbol{w}^{(t)}\|$ appears in numerator and denominator and cancels; then use $(\Sigma \boldsymbol{w})^\top = \boldsymbol{w}^\top \Sigma$ (symmetry of $\Sigma$):

$$E_k(\boldsymbol{w}^{(t+1)}) = \frac{\boldsymbol{w}^{(t+1)\top} \boldsymbol{u}_k}{\boldsymbol{w}^{(t+1)\top} \boldsymbol{u}_1} = \frac{\left(\Sigma \boldsymbol{w}^{(t)} / \|\Sigma \boldsymbol{w}^{(t)}\|\right)^\top \boldsymbol{u}_k}{\left(\Sigma \boldsymbol{w}^{(t)} / \|\Sigma \boldsymbol{w}^{(t)}\|\right)^\top \boldsymbol{u}_1} = \frac{\boldsymbol{w}^{(t)\top} \Sigma \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top} \Sigma \boldsymbol{u}_1}.$$

**(b)** Starting from the result above, show that $E_k(\boldsymbol{w}^{(t+1)}) = |\lambda_k/\lambda_1| \cdot E_k(\boldsymbol{w}^{(t)})$.

**Solution (b):** Use $\Sigma \boldsymbol{u}_i = \lambda_i \boldsymbol{u}_i$:

$$E_k(\boldsymbol{w}^{(t+1)}) = \frac{\boldsymbol{w}^{(t)\top} \lambda_k \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top} \lambda_1 \boldsymbol{u}_1} = \frac{\lambda_k}{\lambda_1} \cdot \frac{\boldsymbol{w}^{(t)\top} \boldsymbol{u}_k}{\boldsymbol{w}^{(t)\top} \boldsymbol{u}_1} = \frac{\lambda_k}{\lambda_1} E_k(\boldsymbol{w}^{(t)}).$$

(Since $\Sigma$ is positive semi-definite, $\lambda_k/\lambda_1 \geq 0$ and equals its absolute value.)

**(c)** Starting from the result above, show that $E_k(\boldsymbol{w}^{(T)}) = |\lambda_k/\lambda_1|^T \cdot E_k(\boldsymbol{w}^{(0)})$, i.e. the convergence of the algorithm is exponential with the number of time steps $T$.

**Solution (c):** Apply (b) repeatedly (induction on $T$; the base case $T = 0$ is trivial):

$$E_k(\boldsymbol{w}^{(T)}) = \frac{\lambda_k}{\lambda_1} E_k(\boldsymbol{w}^{(T-1)}) = \cdots = \left(\frac{\lambda_k}{\lambda_1}\right)^T E_k(\boldsymbol{w}^{(0)}).$$

Because $|\lambda_k/\lambda_1| < 1$ for all $k \geq 2$ (assuming a gap between the largest and the other eigenvalues and $\boldsymbol{w}^{(0)\top} \boldsymbol{u}_1 \neq 0$), every error term decays geometrically, i.e. exponentially in $T$, with rate set by the ratio $|\lambda_2/\lambda_1|$ of the two largest eigenvalues. Thus $\boldsymbol{w}^{(T)}$ becomes orthogonal to $\boldsymbol{u}_2, \ldots, \boldsymbol{u}_d$ relative to its component on $\boldsymbol{u}_1$, i.e. it converges to $\pm \boldsymbol{u}_1$. ■